Mikhail Maslov, PhD

Quantitative Research · Systematic Strategies · Portfolio & Risk Modelling

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Theoretical physicist with experience in mathematical modelling, scientific computing, and machine learning. Seeking quant research roles focused on systematic strategies, portfolio and risk modelling. Willing to relocate Europe-wide.

EDUCATION

PhD in Theoretical Physics, Institute of Science and Technology Austria (ISTA) **(2018–2024)**
BSc in Applied Mathematics and Physics, Moscow Institute of Physics and Technology (MIPT) **(2014–2018)**
Graduated with Honors (GPA 8.6/10), received Abramov and State Academic fellowships.

EXPERIENCE

Publications: 7 publications (+2 under review), 170 citations; reviewer for *Phys. Rev. A* and *Phys. Rev. Research*.

Postdoctoral Researcher, Institute of Science and Technology Austria **(2025–present)**
- Built from the ground up a neural-network solver for a class of few-body quantum differential equations using reinforcement learning and Monte Carlo sampling. Demonstrated accurate convergence to the benchmark DVR solution across broad parameter regimes.
- Developed a symmetry-aware mathematical model for optical sensing of material chirality.

PhD Researcher, Institute of Science and Technology Austria **(2018–2024)**
- Developed a universal group-theoretical framework for analyzing angular momentum exchange in optical systems, applicable to many research fields in physics.
- Formulated analytical models for the experimental control of quantum many-body systems; assessed their feasibility numerically using variational and spectral methods.

Undergraduate Researcher, Moscow Institute of Physics and Technology **(2016–2018)**
- Developed a theoretical framework based on coupled partial differential equations to analyze plasmonic instabilities in nanostructured semiconductor systems.
- Designed and maintained a MATLAB codebase for electrodynamics simulations of two-dimensional electron systems via spectral methods and Fourier analysis.

Intern, Helmholtz Zentrum Dresden-Rossendorf (HZDR) **(Jun–Aug 2017)**
- Designed and optimized code for molecular dynamics simulations on an HPC cluster, using existing Monte Carlo (SRIM) and molecular dynamics (LAMMPS, OpenMPI) codes.

Intern, Deutsches Elektronen-Synchrotron (DESY), Hamburg **(Jun–Aug 2016)**
- Developed MATLAB code for simulating crystallographic data using Monte Carlo methods and comparing it to the experimental data from free-electron laser (XFEL) via Fourier analysis.

Student Representative and Teaching Assistant, Institute of Science and Technology Austria **(2019–2022)**
- Designed curriculum and taught atomic and condensed matter physics to 50+ PhD students
- Managed curriculum and student infrastructure development (500+ PhD students)

Teaching assistant, Moscow Institute of Physics and Technology **(2014–2019)**
- Taught mathematics, physics and computer science to 100+ high-school students

SKILLS

Mathematics: linear algebra, tensor calculus, probability theory, differential equations, mathematical physics, group theory, differential geometry, quantum algorithms, many-body spin systems.

Numerics & ML: Monte Carlo methods, variational methods, finite-difference methods, Fourier analysis, optimization, supervised and reinforcement learning.

Programming: Python (NumPy, SciPy, PyTorch), MATLAB, Wolfram Mathematica.

Computing: vectorization, parallel computing, HPC, Linux, Docker, Bash automation.

Interests: game theory, decision theory, auction theory; market making and risk management.